

TURPAN, China

WMO: 515730

Lat: 42.9503N

Lon: 89.2307E

Elev: 39

StdP: 100.85

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-12.0	-22.8	0.5	-2.9	-21.2	0.6	-5.4	3.4	-5.2	3.0	-5.8	0.5	40	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	42.5	21.3	41.0	20.9	39.6	20.5	23.0	37.9	22.2	36.9	21.6	36.3	1.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.4	29.7	17.1	12.3	30.2	15.9	11.3	31.0	68.6	37.9	65.6	36.9	63.2	36.2	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.2	3.4	-13.9	45.6	2.0	1.5	-15.4	46.7	-16.5	47.5	-17.7	48.4	-19.2	49.5		
(5)			-14.8	25.1	1.8	2.1	-16.1	26.6	-17.1	27.8	-18.1	29.0	-19.5	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	-6.2	2.0	11.9	20.7	26.8	31.9	33.5	31.8	25.5	15.9	5.6	-3.9		
	DBStd	14.21	3.14	3.99	4.89	4.39	3.50	3.04	2.76	2.90	3.66	4.25	4.09	3.53		
	HDD10.0	1338	502	224	37	1	0	0	0	0	0	4	139	432		
	HDD18.3	2627	760	457	207	27	1	0	0	0	2	101	383	690		
	CDD10.0	3655	0	1	94	322	521	656	728	676	466	186	7	0		
	CDD18.3	1902	0	0	6	98	263	406	469	418	218	24	0	0		
	CDH23.3	26553	0	0	52	983	3202	6146	7441	6182	2368	179	0	0		
	CDH26.7	16273	0	0	6	374	1693	3982	5064	3965	1151	38	0	0		
(14) Wind	WSAvg	1.2	0.7	0.9	1.3	1.5	1.6	1.6	1.5	1.4	1.2	0.9	0.7	0.7		
(15) Precipitation	PrecAvg	16	1	0	2	0	1	3	2	2	2	1	1	1		
	PrecMax	37	10	6	26	2	10	17	14	10	12	11	13	8		
	PrecMin	3	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	9	2	1	5	1	3	4	3	2	3	3	3	2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	14.1	27.4	35.3	39.7	43.4	45.1	43.4	38.4	30.1	18.1	6.1		
		MCWB	-1.5	5.5	12.3	16.4	18.9	20.7	21.9	21.3	19.7	15.7	9.0	1.9		
	2%	DB	1.0	11.7	24.7	32.6	37.7	41.5	43.1	41.5	36.2	27.4	15.7	3.8		
		MCWB	-2.7	4.0	11.0	15.4	18.2	20.4	21.9	21.0	18.9	14.6	7.7	-0.1		
	5%	DB	-0.4	10.1	22.4	30.7	35.9	40.1	41.4	40.0	34.4	25.3	13.8	2.2		
		MCWB	-3.6	3.0	10.0	14.6	17.8	20.0	21.4	20.8	18.3	13.5	6.7	-1.2		
	10%	DB	-1.6	8.4	20.0	28.7	34.1	38.5	39.9	38.4	32.6	23.1	11.9	0.8		
		MCWB	-4.5	2.0	8.8	13.7	17.1	19.8	21.1	20.5	17.5	12.5	5.7	-2.1		
	0.4%	WB	-1.0	5.8	12.8	17.1	20.6	22.7	24.7	23.7	21.5	16.4	9.7	2.2		
		MCDB	2.2	13.4	26.1	32.8	36.2	37.7	38.8	36.7	34.4	28.1	16.7	5.7		
	2%	WB	-2.4	4.3	11.5	15.9	19.2	21.8	23.4	22.6	20.1	15.2	8.3	0.4		
		MCDB	0.6	11.1	23.7	31.4	35.3	36.6	38.5	36.5	33.2	25.9	14.6	3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	-3.4	3.2	10.2	15.0	18.4	21.1	22.6	21.9	19.0	14.1	7.1	-0.8		
		MCDB	-0.6	9.7	21.9	29.5	34.1	36.2	37.8	36.2	31.8	24.0	13.1	2.0		
	10%	WB	-4.4	2.1	9.0	14.1	17.6	20.5	21.9	21.2	18.1	13.0	6.0	-2.1		
		MCDB	-1.7	8.1	19.6	27.6	32.7	35.8	36.8	35.7	31.2	22.3	11.5	0.8		
	5% DB	MDBR	7.0	9.2	10.7	11.9	12.2	12.0	11.8	12.6	12.9	12.2	9.1	6.1		
		MCDBR	7.7	10.4	12.6	14.4	14.8	13.9	14.1	14.5	14.3	13.9	11.4	6.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.5	6.1	5.9	5.8	5.7	4.7	4.7	4.8	5.1	5.5	5.9	4.5		
		MCDBR	7.5	10.3	12.1	13.4	13.4	12.3	12.4	13.1	13.5	13.6	11.3	6.3		
	5% WB	MCWBR	5.4	6.1	5.7	5.4	5.5	4.6	4.8	4.9	5.2	5.3	5.8	4.4		
		MCDBR	7.5	10.3	12.1	13.4	13.4	12.3	12.4	13.1	13.5	13.6	11.3	6.3		
(40) Clear-Sky Solar Irradiance	taub		0.418	0.473	0.540	0.585	0.583	0.588	0.564	0.545	0.527	0.503	0.465	0.420		
	taud		1.917	1.827	1.704	1.667	1.719	1.779	1.868	1.890	1.859	1.841	1.883	1.930		
	Ebn at Noon		648	679	687	698	717	715	728	724	694	639	589	599		
	Edh at Noon		127	165	211	235	229	217	196	186	178	160	131	116		
(44) All-Sky Solar Radiation	RadAvg		2.24	3.25	4.56	5.75	6.70	6.86	6.51	6.06	5.23	3.79	2.49	1.93		
	RadStd		0.13	0.16	0.20	0.36	0.28	0.43	0.33	0.34	0.21	0.17	0.14	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.81	N/A	N/A	+1.04	-0.58	-1.60	N/A	N/A	+298	+219
(47) Regional (3 neighbors)		+0.47	N/A	N/A	N/A	N/A	-0.83	N/A	N/A	+131	+90

Nomenclature: See separate page